ROOT CAUSE UNLOCKED · ULTIMATE GUIDE

Unlock Your *Energy*, Ultimate Guide

Why You're Exhausted, Why "Just Sleep More" Never Fixed It, and What Your Cells Are Actually Asking For

A guide for anyone living with unexplained fatigue, fibromyalgia, chronic fatigue syndrome (ME/CFS), or insomnia who has been told their labs look normal and has not gotten a straight answer since.

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BEFORE YOU START

> ***Disclaimer:*** *This book is for educational purposes only and does not constitute medical advice. It does not diagnose any condition, and reading it does not make you my patient. Always consult a qualified healthcare provider before starting any new treatment, supplement, or dietary change, and do not start, stop, or change any prescribed medication without speaking to the prescriber. If you have myalgic encephalomyelitis, chronic fatigue syndrome, or post-exertional malaise, review any exercise or activity recommendation with a practitioner familiar with post-exertional malaise before you start it.*

A note before you start. This guide is educational information about the biology behind chronic fatigue, fibromyalgia, and insomnia, and the general categories of intervention used to address the underlying drivers. It is not individualized medical advice, it does not diagnose any condition, and it does not replace a relationship with your physician or a clinical evaluation with me and my team. If you take a prescription medication, keep taking it exactly as prescribed, and talk with your prescriber before changing anything. If you are new to a supplement, pregnant or nursing, or you take blood thinners, thyroid medication, or other prescriptions, talk to a qualified practitioner first.

A safety note before anything else. New or worsening fatigue that comes with unexplained weight loss, fever, night sweats, fainting, chest pain, shortness of breath, or any new neurological symptom (weakness on one side of the body, vision changes, slurred speech, sudden confusion) is not something to address with a supplement guide. It warrants prompt medical evaluation. This guide is written for the common, longstanding pattern of unexplained fatigue, fibromyalgia, and insomnia once your physician has ruled out an acute or dangerous cause, not as a replacement for that first workup.

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INTRODUCTION

The Tiredness That Doesn't Add Up

You are getting seven or eight hours in bed most nights. Your labs came back "normal." Your doctor was not dismissive, exactly, but there was not much else offered beyond "try to manage your stress" or a suggestion to consider an antidepressant. And you are still exhausted in a way that a good night's sleep does not touch, that a second cup of coffee does not touch, that makes ordinary life (the school run, the work day, the grocery list) feel like wading through wet sand.

If this is you, or if you have been told your exhaustion is fibromyalgia, or that your labs are borderline, or that you just are not sleeping well and should try harder to relax, you already know the frustrating part. Fatigue is the single most common complaint walked into a functional medicine practice, and it is also the complaint most likely to be waved off because nothing on a standard panel looks alarming enough to act on.

Here is the reframe this guide is built around. Fatigue, chronic pain that will not resolve, and sleep that does not restore you are rarely a single problem with a single fix. They are usually the downstream signal of several converging drivers: how well your mitochondria are converting fuel into usable cellular energy, how your stress response system (the HPA axis) is regulating cortisol, whether your sleep architecture is actually doing its overnight repair work, what is happening in your gut, whether specific nutrients your energy systems depend on are actually where they need to be, and how your thyroid and blood sugar regulation are holding up under all of it. None of these show up clearly on a basic metabolic panel and a single TSH. All of them are testable, and all of them are addressable.

This guide walks through eight keys: the cellular energy story, the HPA axis, sleep, the gut, nutrient status, thyroid and blood sugar, the specific and important case of post-exertional malaise if you have chronic fatigue syndrome or fibromyalgia, and what functional testing actually looks for. If you have been diagnosed with (or suspect) myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) or fibromyalgia, Key 7 matters more than any other chapter in this book. Some of the advice that helps garden-variety fatigue is specifically wrong, and potentially harmful, for you, and that chapter explains why and what to do instead.

One more thing before we start. If you have also been dealing with irregular cycles, low libido, or other signs your sex hormones are out of balance, that is real and it is worth pursuing, and it is not what this guide is about. This guide covers the HPA axis and cortisol in depth because stress physiology touches nearly everything downstream of it, but it stays in its own lane on sex hormone testing and treatment. Raise that side of the picture with me or with your physician, because it deserves a proper look rather than a paragraph here.

KEY 1

Your Fatigue Might Be a Cellular Problem, Not a Character Flaw

The part nobody mentions. Fatigue gets treated as a mood, a discipline issue, or a sleep hygiene problem, something you should be able to push through with enough coffee and willpower. For a large number of people living with persistent, unexplained fatigue, that framing is backwards. The exhaustion is a downstream signal of how efficiently your cells are actually producing energy, and when that production line is impaired, willpower does not fix it.

What's actually happening, in plain terms. Every cell in your body converts the food you eat and the oxygen you breathe into ATP, the actual energy currency your muscles, brain, and organs run on, inside structures called mitochondria. That conversion happens in stages: glycolysis breaks down glucose, the Krebs cycle processes the result into intermediates, and the electron transport chain uses those intermediates to generate the bulk of your ATP. Each stage depends on specific nutrient cofactors (B vitamins, magnesium, CoQ10, alpha lipoic acid, iron) to keep running smoothly. When any of those cofactors are short, or when the mitochondria themselves are under oxidative stress, the whole chain backs up, and the result is measurable, not imagined. In fibromyalgia specifically, researchers have documented oxidative stress and mitochondrial dysfunction directly, including reduced CoQ10 in blood cells, a compound your mitochondria depend on as both an antioxidant and an electron carrier in that final energy producing stage.[1] In myalgic encephalomyelitis and chronic fatigue syndrome (ME/CFS), newer research using muscle biopsies and imaging has found actual structural damage to mitochondria inside skeletal muscle tissue, along with a pattern of sodium and calcium overload in muscle cells that appears to both cause and worsen with exertion.[2] This is a meaningfully different picture than simply being out of shape or deconditioned.

What actually helps, at an educational level. The general categories that support this system are the ones your Krebs cycle and electron transport chain actually run on: B vitamins (particularly B1, B2, B3, and B12), magnesium, CoQ10 (the ubiquinol form is more readily used by the body), alpha lipoic acid, and adequate iron and protein to build the enzyme machinery in the first place. None of these work as a single silver bullet, and dosing them without knowing which one you are actually short on is guesswork, which is exactly the case Keys 5 and 8 make for testing rather than assuming. What matters here is the reframe: if you have been told your labs are normal and your fatigue is “just” stress or deconditioning, mitochondrial and cellular energy status is a real, measurable layer that a standard metabolic panel was never designed to look at.

KEY 2

The HPA Axis, Cortisol, and Why "Just Relax" Misses the Point

The piece that gets oversimplified. Cortisol gets a bad reputation as simply "the stress hormone," something to minimize. In reality, cortisol on a healthy rhythm (higher in the morning to help you wake and mobilize, tapering across the day, low at night so you can sleep) is one of the most important permissive hormones in the body. It helps regulate blood sugar, inflammation, and blood pressure, and it partners with your thyroid to keep the broader endocrine system working together. When the hypothalamic-pituitary-adrenal (HPA) axis, the signaling loop between your brain and your adrenal glands that produces this rhythm, gets dysregulated, the fallout goes well beyond feeling stressed.

The mechanism, unpacked. In chronic fatigue syndrome specifically, a substantial body of research has documented a consistent pattern of HPA axis change: mild hypocortisolism (lower than typical cortisol output), a flattened diurnal curve (less of that healthy morning-high, evening-low rhythm), heightened negative feedback, and a blunted response when the system is challenged.[3] These changes are clinically meaningful. They correlate with worse symptoms and poorer response to standard treatment, and women with CFS are more likely than men to show reduced cortisol specifically.[3] It is worth being precise about what this does and does not mean. The honest picture is more nuanced than "broken adrenals": some research suggests HPA axis changes are not present in the earliest stages of these illnesses, and may develop as a consequence of prolonged inactivity, deconditioning, and disrupted sleep rather than causing the illness outright, while other factors (low activity, depression, early life stress) independently push cortisol down as well.[4] What is clear is that once the HPA axis is dysregulated, it becomes part of what perpetuates the fatigue, whichever direction the original arrow pointed. And cortisol's reach extends further than most people realize. It also suppresses the enzyme that converts inactive thyroid hormone (T4) into its active form (T3), one reason chronic stress and thyroid-driven fatigue so often travel together, a connection Key 6 returns to.[5]

Where to start, at an educational level. The starting point is not another adaptogen picked off a supplement store shelf. It is finding out where your own cortisol curve actually sits, which requires a multi-point test across the day (a single blood draw at nine in the morning tells you almost nothing about a rhythm), because a flat, low curve and a spiked, "wired and tired" curve are different problems that call for different support. Sleep, discussed next in Key 3, is not a lifestyle add-on here. It is one of the most direct levers on HPA axis regulation available to you, alongside nervous system regulation practices (paced breathing, time outdoors, reducing unnecessary stimulant load) that calm the upstream signal rather than just propping up the downstream hormone. If you are also dealing with irregular cycles, low libido, or other sex hormone symptoms, that is worth a separate, dedicated look with a clinician, and appropriate sex hormone testing is a reasonable next step alongside the stress-axis work this chapter describes.

KEY 3

Why Eight Hours in Bed Does Not Always Mean Eight Hours of Repair

The complaint that confuses people. "I'm in bed for eight hours and I still wake up exhausted" is one of the most common sentences said in a functional medicine intake, and it confuses people because it seems to contradict the basic sleep math. The missing piece is that sleep is not one uniform state. It cycles through stages, and the deep, slow-wave stages, where the bulk of physical repair and immune housekeeping happens, can be present in name only if they are shallow, fragmented, or crowded out by lighter stages. Fibromyalgia in particular is strongly associated with this kind of non-restorative sleep, where total time in bed looks adequate but the architecture underneath it is disrupted.

How this actually works. Sleep and your immune and inflammatory systems are bidirectionally linked, not a one-way relationship where poor health simply causes poor sleep. Adequate sleep supports immune function and helps keep inflammatory signaling in check, while prolonged sleep deficiency (whether that means short duration or fragmented, non-restorative sleep) drives chronic, low-grade systemic inflammation, the same inflammatory backdrop that shows up across the fatigue and pain conditions this guide addresses.[6] That is a two-way street. Poor sleep raises inflammation, and the inflammatory and HPA axis disruption already covered in Key 2 makes sleep less restorative in turn, which is exactly the kind of self-reinforcing loop that makes "just sleep more" unhelpful advice on its own.

What actually moves the needle, at an educational level. Sleeping pills address the symptom of not falling asleep. They do very little for sleep architecture itself, and long-term use carries its own tradeoffs worth discussing with your prescriber. The single most well-evidenced non-pharmacological approach for chronic insomnia is cognitive behavioral therapy for insomnia (CBT-I), a structured, short-term program (not a single relaxation tip) covering sleep restriction, stimulus control, and cognitive strategies around sleep anxiety. A major systematic review and meta-analysis found meaningful, sustained improvements across objective and diary-based sleep measures, including sleep efficiency and time spent awake after falling asleep, with no adverse outcomes reported.[7] Alongside CBT-I, magnesium (covered further in Key 5), consistent light exposure timing, and addressing the HPA axis and gut drivers covered elsewhere in this guide all support the underlying architecture rather than just knocking you out for the night. If insomnia has been part of your picture for months or years, treating it as its own priority, not just a symptom that will resolve once "the real problem" is fixed, is often exactly backwards: unrestored sleep is frequently one of the real problems.

KEY 4

The Gut-Energy Connection

The disconnect nobody points out. It is easy to file digestive symptoms and fatigue as two separate complaints on an intake form. They are frequently the same problem wearing two different costumes. Your gut lining, the trillions of microbes living along it, and your immune system are in constant communication, and when that ecosystem is disrupted, the consequences reach well past bloating and irregularity.

The biology behind it, in plain terms. Research specifically in ME/CFS patients has found reduced diversity in the gut microbiome compared to healthy controls, along with a shift toward more pro-inflammatory and fewer anti-inflammatory bacterial species. That same research found elevated blood markers of microbial translocation, meaning bacterial components (lipopolysaccharide and related markers) were more likely to be crossing from the gut into circulation than in healthy controls, a pattern that correlated with the inflammatory picture seen in these patients.[8] This matters for energy specifically because that translocation, and the low-grade inflammation it drives, places an ongoing burden on the same mitochondrial and HPA axis systems covered in Keys 1 and 2. Your gut is not a bystander to your energy levels. It is one of the upstream inputs.

The approach that works, at an educational level. This is not a case for a generic probiotic and calling it done. The useful sequence, the same one used for any gut repair approach, is removing what is actively feeding dysbiosis or inflammation (food triggers, unnecessary medications that damage the gut lining, unaddressed infections), replacing what is missing (stomach acid, digestive enzymes, bile support if needed), reinoculating with a microbiome that stool or breath testing actually indicates you need rather than guessing, and repairing the gut lining itself (nutrients such as zinc carnosine, glutamine, and butyrate support this stage). Gut testing, discussed further in Key 8, is what turns this from a guess into a targeted plan, and it is a reasonable place to start if digestive symptoms, however minor, have been part of your picture alongside the fatigue.

KEY 5

Nutrient and Mineral Status: Iron, B12, Magnesium, and CoQ10

The gap between two different statements. “Your labs are normal” and “your nutrient status is optimal for energy production” are not the same statement, and the gap between them is where a lot of unexplained fatigue lives. Standard panels typically check whether you are anemic or grossly deficient. They do not typically check whether your ferritin, your active B12, or your magnesium and CoQ10 status are actually where they need to be to run the energy pathways covered in Key 1.

What the research actually shows. A comprehensive review of the biochemical and clinical evidence on vitamins and minerals for energy and fatigue highlights B vitamins, vitamin C, iron, magnesium, and zinc as having genuinely established roles in energy yielding metabolism, oxygen transport, and the neuronal function that underlies both physical and mental fatigue.[9] A few of these deserve specific mention because they are so commonly overlooked. Iron deficiency without full-blown anemia is a real, well-studied cause of fatigue. A randomized controlled trial in menstruating women with a ferritin under 50 (a level many labs still call “normal”) and no anemia found that fatigue scores improved significantly more with oral iron than with placebo over twelve weeks, roughly a 48 percent reduction versus 29 percent with placebo.[10] Magnesium is involved in more than three hundred enzymatic reactions, including every step of ATP synthesis, and a dedicated literature review on magnesium and fibromyalgia found consistent associations between low magnesium status and fibromyalgia symptoms across the studies examined, alongside trials testing magnesium supplementation for symptom relief.[11] CoQ10 is worth naming specifically because of its role in the final stage of mitochondrial energy production (Key 1) and because fibromyalgia patients have been shown to run measurably low on it (Key 1). A randomized, placebo-controlled trial combining CoQ10, magnesium, and tryptophan in fibromyalgia patients found significant improvement in pain, sleep quality, and overall functional impact scores after three months, though the fatigue-specific measure in that particular trial did not separate significantly from placebo, an honest result worth knowing rather than a reason to dismiss the approach outright.[12]

What to prioritize, at an educational level. The pattern across all of these nutrients is the same: guessing is expensive and imprecise, and testing is not. Serum levels reflect what has been circulating in roughly the past week. Cellular levels, measured inside your white and red blood cells, reflect what your tissues have actually had available over the past few weeks to months, and the two numbers do not always agree. A ferritin that is technically “in range” can still be too low for optimal energy production. A serum B12 that looks fine can mask a cellular or functional deficiency, particularly if methylmalonic acid, a marker that rises specifically when B12 is not doing its job at the cellular level, is elevated. This is exactly the gap appropriate functional testing, covered in Key 8, is built to close.

KEY 6

Thyroid and Blood Sugar: The Metabolic Thermostat

The blind spot in both systems. Your thyroid sets the metabolic pace for essentially every cell in your body, and your blood sugar regulation determines how steady your fuel supply actually is across the day. When either one drifts, even mildly, fatigue is often the first and most prominent symptom, and both are prone to a specific blind spot: results that fall inside a “normal” reference range while still being functionally suboptimal for you.

The mechanism behind the overlap. Subclinical hypothyroidism (an elevated TSH with normal thyroid hormone levels) affects up to one in ten adults, and middle-aged patients with this pattern commonly report fatigue, cognitive fog, and altered mood even though a standard panel reads as technically within range.[13] As Key 2 touched on, chronic HPA axis activation and elevated cortisol do not just affect your stress response. They suppress the enzyme that converts inactive T4 into active T3, promote insulin resistance and visceral fat storage, and shift the whole metabolic picture toward the pattern sometimes called metabolic syndrome: abdominal weight gain, elevated blood pressure, and impaired blood sugar regulation.[5] This is not a coincidence limited to a few unlucky patients. Fibromyalgia and metabolic syndrome show meaningful overlap in the research, sharing chronic low-grade inflammation, autonomic nervous system dysfunction, and HPA axis dysregulation as common underlying threads, which means the thyroid, blood sugar, and stress-hormone pictures in this guide are not really separate chapters so much as different windows into the same underlying terrain.[14] Blood sugar swings (spikes followed by crashes) place their own direct demand on this system: every crash triggers a cortisol and adrenaline response to bring glucose back up, one more input into the HPA axis dysregulation already covered in Key 2.

The next step, at an educational level. A TSH alone is a screening test, not a complete thyroid picture. Free T3 (the active hormone your cells actually use) and reverse T3 (an inactive form that rises under stress, illness, and caloric restriction, and can effectively block T3 from doing its job even when free T3 looks adequate) fill in a picture a TSH cannot. On the blood sugar side, a fasting glucose and A1c are useful but relatively late-stage indicators. Fasting insulin tends to catch a resistance pattern years before glucose itself drifts out of range. None of this replaces your physician’s thyroid management, especially if you are already on thyroid medication. It is about seeing the fuller picture underneath a “normal” TSH so the right conversation can happen with the right data.

KEY 7

Post-Exertional Malaise and Pacing (Read This One Especially If You Have ME/CFS or Fibromyalgia)

The part where good advice turns wrong. If you have been diagnosed with, or suspect, myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) or fibromyalgia, this is the single most important chapter in this guide, because some well-meaning, generically reasonable advice for ordinary fatigue is specifically wrong for you. Post-exertional malaise (PEM) is a hallmark, defining feature of ME/CFS and is also common in fibromyalgia: a delayed, disproportionate worsening of symptoms (fatigue, pain, cognitive function, and more) that follows physical, cognitive, or even emotional exertion, often twenty-four to seventy-two hours later rather than immediately. A systematic review and meta-analysis pooling fifteen studies found a small to moderate but real and measurable increase in pain following standardized exercise testing in ME/CFS and fibromyalgia patients compared to healthy controls, and critically, the effect was larger when pain was measured eight to seventy-two hours after exercise than when measured immediately afterward, exactly the delayed pattern patients describe and that clinicians sometimes miss because they are not looking for it days later.[15]

The biology behind why this is different. This is not deconditioning, and it is not something you can train through. Emerging research using muscle biopsies and imaging has found direct evidence of mitochondrial damage inside skeletal muscle tissue in ME/CFS patients, along with a pattern of sodium and calcium overload following exertion that appears to drive further mitochondrial injury, creating a cycle where exertion itself causes measurable cellular damage rather than the ordinary, adaptive muscle stress that builds fitness in a healthy person.[2] This is the mechanistic reason a well-intentioned “just push through it and build back your stamina” approach can backfire badly in this population.

What to do about it, at an educational level, stated plainly. Graded exercise therapy (GET), the structured, progressively increasing exercise programs sometimes still recommended for chronic fatigue, is not appropriate for patients with PEM, and this guide does not recommend it. This is not a matter of taste. A review of the evidence found that outside of controlled trial conditions, many ME/CFS patients report genuine deterioration with graded exercise therapy, that exercise physiology studies reveal real abnormalities in how these patients respond to exertion, and that the safety of these programs cannot be assumed based on the available evidence.[16] A separate, detailed review concluded plainly that graded exercise therapy is not only unproven as an effective treatment for ME/CFS but is reported by many patients to cause substantial deterioration, worsening the same inflammatory, immune, and HPA axis abnormalities that already underlie the illness.[17] If you have ME/CFS or PEM-predominant fibromyalgia, and someone (including a well-meaning trainer, family member, or even a clinician unfamiliar with PEM) suggests pushing through fatigue to build tolerance, that advice does not apply to you, and following it risks a genuine setback, not just a hard workout.

The alternative is pacing: learning and staying inside your “energy envelope,” the amount of physical, cognitive, and emotional activity you can do on a given day without triggering a PEM crash the following day or two. Practically, this looks like breaking activities into shorter segments with rest between them, tracking your own patterns (the worksheet at the end of this guide includes a simple version of this), and treating a good day as information, not permission to make up for lost time by doing twice as much. Pacing is not giving up. It is working with a system that has a measurably different response to exertion than an average, healthy nervous and mitochondrial system does, and it is the approach with an honest evidence base behind it, rather than the one that sounds more motivating on a poster.

If you do not have PEM, ordinary graded reconditioning after a period of inactivity is a different, reasonable conversation to have with your practitioner. The distinction matters enough that it is worth confirming which situation you are actually in before starting any exercise program, which is part of what appropriate testing and evaluation (Key 8) is for.

KEY 8

What Appropriate Functional Testing Actually Looks Like

The frustrating half of the truth. “We ran your labs and everything came back normal” is one of the most common, and most frustrating, sentences in chronic fatigue care, and it is often true and also incomplete. A standard panel (a CBC, a basic metabolic panel, maybe a TSH) is a reasonable first pass and rules out some serious causes, but it was never designed to evaluate mitochondrial function, HPA axis rhythm, cellular nutrient status, or gut microbial balance, the systems this guide has walked through.

What’s really going on underneath. There is a real diagnostic challenge underneath this, worth naming honestly. A formal review of diagnostic methods for ME/CFS found that nine different sets of clinical criteria have been used to define the condition over the years, and concluded that none of the current diagnostic methods have been thoroughly validated for identifying patients when real diagnostic uncertainty exists.[18] Fibromyalgia, by contrast, has a more settled diagnostic picture: the current criteria combine a widespread pain index and a symptom severity scale, refined and validated specifically to reduce misclassification against other regional pain conditions.[19] In both cases, the diagnosis itself is a clinical judgment built from history and pattern, which is exactly why the testing that follows a diagnosis, or a strong suspicion, needs to be purposeful rather than a shotgun panel.

What appropriate testing actually looks like, at an educational level. A useful evaluation for persistent fatigue, fibromyalgia, or insomnia typically layers a few categories rather than ordering everything at once. Blood chemistry and inflammatory markers (a CBC with differential, a comprehensive metabolic panel, ferritin and iron studies, hs-CRP) establish the baseline and catch the more common, more treatable drivers, like the iron and thyroid patterns covered in Keys 5 and 6, first. A four-point diurnal cortisol test (morning, midday, afternoon, and evening, rather than a single blood draw) is what actually shows the HPA axis pattern described in Key 2, whether that is a flat, low curve or a wired, high one, because those two patterns call for different approaches. Organic acids testing looks at urine markers of how efficiently your Krebs cycle and electron transport chain are actually running, alongside markers of B vitamin status, oxidative stress, and gut microbial byproducts, in a single sample, which makes it a reasonable broad first look when the picture is complex or multi-system. A comprehensive micronutrient panel goes a step further than a basic vitamin blood draw by measuring nutrient status inside your white and red blood cells, not just what is circulating that week, which is the more clinically meaningful number for the reasons Key 5 described. Depending on your history, more targeted testing (a mycotoxin panel if mold exposure is suspected, viral titers if a post-viral onset is suspected, a comprehensive stool or breath test if gut symptoms are prominent) may be added, but the sequencing matters: broad screening first, then targeted follow-up aimed at whatever the pattern actually points to.

One honest caveat belongs here. A recent systematic review of dietary supplement trials specifically in ME/CFS found that while a handful of interventions showed promising signals for fatigue, the overall body of evidence carries a high risk of bias, small sample sizes, and inconsistent results, and does not yet support firm, one-size-fits-all conclusions.[20] That is precisely the case for testing rather than guessing, and for working with a practitioner who will tell you plainly when the evidence is still catching up rather than overselling certainty that is not there yet.

A Staged Roadmap: An Order of Operations, Not a Schedule

Chronic fatigue, fibromyalgia, and insomnia did not develop overnight. What I can give you honestly is an order of operations, not a set of dates, because I have no way to know from here how fast your system moves. Plateaus and occasional flares are a normal part of the process rather than a sign that something has failed.

I am deliberately not attaching timelines to outcomes. Anyone who tells you what percentage better you will be, and by when, is guessing with your money. What replaces that is your own log. Two weeks of baseline scores before you change anything, then the same two numbers most days, and every judgment made against your own starting point rather than against a promise. That is the only honest measuring stick in this condition, and it is also the more useful one.

First, the foundation. Sleep architecture, nervous system regulation, and the most correctable nutrient gaps (iron, B12, magnesium) if testing identifies them. These come first because they are the cheapest to run, the safest to be wrong about, and the ones that make everything after them work better.

Then the systems that reinforce each other. Gut repair, stress-axis support, and mitochondrial foundation work tend to happen together rather than in strict sequence, since each one takes load off the others.

Then the deeper drivers, if testing found any. Thyroid, post-viral pictures, mold or toxin exposure, gut infections. These get dedicated attention once the foundation is holding, not before, because working on them through a broken sleep pattern wastes the effort.

And throughout, pacing. If post-exertional malaise is part of your picture (Key 7), it governs the whole plan rather than sitting inside it. In my clinical experience the people who stall earliest are usually the ones who moved faster than their own system could absorb, not the ones who took the sequence seriously. Judge in months and seasons, against your log, not against how you feel on any given Tuesday.

Your Energy and Symptom Tracker

Bring this filled out, or a version of it, to your next appointment. Patterns across two to four weeks are far more useful to a practitioner than a single bad day or a single good day.

Daily tracker

Recreate this as a simple table for each day of the week, or photocopy it.

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Energy on waking (0 to 10)							
Energy by mid-afternoon (0 to 10)							
Hours slept							
Woke feeling rested? (Y/N)							
Pain level, if applicable (0 to 10)							
Brain fog (0 to 10)							
Did today include a PEM crash? (Y/N)							
First meal of the day							
Caffeine (count)							
Notable stressors							

The energy envelope worksheet

Especially useful if you have ME/CFS, fibromyalgia, or suspect post-exertional malaise. Revisit Key 7 before filling this out.

1. On an average, non-crash day, list the activities you can reliably do without triggering a delayed worsening: __________
2. List activities that reliably trigger a crash, even a mild one, and roughly how many hours or days later it shows up: __________

3. This week, pick one borderline activity and break it into two shorter segments with a rest period between them, rather than doing it all at once. Note what happened: __________
4. On your best days this week, did you do meaningfully more than usual to “catch up”? If yes, what happened over the following one to two days? __________
5. One thing you will pace differently next week: __________

Questions worth bringing to your practitioner

- Has anyone actually run a four-point diurnal cortisol test, or just a single morning draw?
- What was my ferritin number specifically, not just whether it was “in range”?
- Has my free T3 and reverse T3 been checked, or only TSH?
- If I have ME/CFS or PEM-predominant fibromyalgia, does my current exercise or activity recommendation account for that, or is it a generic reconditioning plan?

CONCLUSION

You Were Not Imagining This

If you came to this guide after being told your labs looked fine, that you should try to relax more, or that fatigue is just something to manage indefinitely, the point of everything above is simple: there is real, measurable biology underneath persistent fatigue, fibromyalgia, and insomnia, and most of it sits just outside what a standard panel checks. Mitochondrial function, HPA axis rhythm, sleep architecture, gut health, cellular nutrient status, thyroid and blood sugar regulation, and, for a meaningful subset of you, post-exertional malaise, are all real, testable, addressable layers.

None of this is a promise of a specific outcome, and it is not a substitute for working with a practitioner who can look at your actual labs, history, and pattern. What it is meant to be is a map: a way to walk into your next appointment asking sharper questions, and a way to understand that “we don’t know why you’re so tired” usually means the right systems have not been examined yet, not that there is nothing to find.

If you would like help turning this into a personalized, tested plan rather than working through it alone, that is exactly the kind of evaluation my team and I do at Seay Wellness every day. Bring your tracker, bring your questions, and bring whatever labs you already have. The rest is a conversation, not a sales pitch.

Take the Next Step: Your Free Root Discovery Session

Everything in this guide is meant to help you understand your own picture and walk into a clinical conversation with better questions, not to replace that conversation. The Root Discovery Session is a complimentary consultation with my team to review your fatigue timeline, your sleep, what has already been tested, and which of the drivers in this book have never actually been looked at. There is no obligation, and if you decide to move forward, this conversation is simply the first step.

Schedule your free Root Discovery Session at rootcauseunlocked.com/discovery.

You can also reach the office directly at (919) 662-0520.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

If you would rather start on your own, start tracking.

The tracker earlier in this guide works on paper. The digital version does the same job on your phone, keeps your history in one place, and shows you the trends that matter here: which days cost you, how sleep is actually tracking against energy, and whether a crash follows exertion on a delay. Two weeks of honest data makes any future appointment, with our team or your own physician, dramatically more useful than memory alone.

Start your free tracker at rootcauseunlocked.com/tracker.

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Disclaimer

This guide provides general educational information about the biology of chronic fatigue, fibromyalgia, and insomnia. It is not a substitute for professional medical advice, diagnosis, or treatment, and nothing in it should be used to self-diagnose or self-treat. Always consult a qualified healthcare provider before starting, stopping, or changing any supplement, medication, or exercise program, particularly if you are pregnant, nursing, managing another medical condition, or taking prescription medications. Never stop or change a prescription medication based on anything in this guide; work with the prescribing provider. If you have myalgic encephalomyelitis/chronic fatigue syndrome or post-exertional malaise, review any exercise or activity recommendation with a practitioner familiar with PEM before starting it. New or worsening fatigue accompanied by unexplained weight loss, fever, night sweats, fainting, chest pain, shortness of breath, or new neurological symptoms warrants prompt medical evaluation rather than a supplement or lifestyle approach. This guide is for education and screening purposes only and does not diagnose any condition.

References

Every citation below was independently retrieved and confirmed against its live PubMed record (esearch, esummary, and a full abstract via efetch) before use. Full verification detail, including a claim from an internal source document that was checked and could not be verified, is in the companion file Energy-Citation-Ledger.md.

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From Understanding to Doing

Everything up to this point was about understanding: what is happening in your body, why it may be happening, and which directions are worth investigating. That is the half a book can do well.

What follows is the other half, and it is a different kind of document on purpose. Three guides, written to be used rather than read straight through. The first tells you which lab panels answer which question and what the numbers mean, and it includes a request list you can hand to a clinician. The second covers what to eat and, more usefully, how to test a dietary change on yourself so you end up with an answer instead of a habit. The third walks through what has actually been studied, at what dose, and names the products whose trials came back negative.

Each one stands alone, so use them in whatever order matches what you are stuck on. If you are not sure where to begin, begin with the lab guide, because knowing what is driving your symptoms changes what you should do about the other two.

A note on repetition. A few points appear in both halves. That is deliberate. The ones repeated are the ones where getting it wrong causes harm, and they are worth meeting twice.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

See a doctor promptly, ahead of any of this, if your fatigue comes with: unintentional weight loss, fever or night sweats, a new lump, blood in stool or urine, coughing blood, chest pain, breathlessness at rest or on mild exertion, fainting, new confusion, or severe headache. New fatigue in someone over 60, or fatigue that has worsened steadily over weeks rather than fluctuating, also deserves a medical workup first. Fatigue is a symptom of a great many things, and some of them are urgent.

If you have thoughts of harming yourself, please call or text 988 in the United States for the Suicide and Crisis Lifeline, or go to an emergency department.

Read this before you read anything else about exercise

If your symptoms reliably get **worse a day or two after exertion**, and that worsening is out of proportion to what you did, that is post-exertional malaise. It is the defining feature of myalgic encephalomyelitis and chronic fatigue syndrome, and it changes the entire plan.

Where post-exertional malaise is present, graded exercise is not the answer, and pushing through causes harm. Key 7 of *Unlock Your Energy* covers this and it is the most important safety line in the whole kit. Everything in these guides is written to be compatible with pacing rather than with pushing.

If that describes you, read the pacing section of the supplement guide before you change anything.

The eight keys, and what tests each one

Unlock Your Energy works through eight root causes. Blood work answers some of them cleanly, and for others the honest answer is that no panel will tell you.

Key	What it means	How you find it
1. Cellular	Mitochondrial energy production	No validated clinical test. Inferred, not measured
2. HPA axis	The stress response system and cortisol rhythm	Contested. See the honest section below
3. Sleep	Hours in bed are not hours of repair	Sleep study, and a two-week sleep log. Not a blood test
4. Gut	Absorption, and the microbial contribution	Celiac serology, stool analysis where symptoms justify it
5. Nutrients	Iron, B12, folate, magnesium, vitamin D	**Blood work, and this is where the clearest answers live**
6. Metabolic	Thyroid and blood sugar	TSH with free T4 and free T3, fasting glucose, HbA1c
7. Pacing	Post-exertional malaise and the energy envelope	History. The most consequential finding in this guide has no test
8. Testing	Choosing the right panel rather than the biggest one	This guide

Notice the shape of that. The two most decisive items, post-exertional malaise and sleep quality, are not blood tests. The nutrient key is the one where a lab draw genuinely changes what you do, which is why most of this guide is about it.

Key 5: nutrients, where the clearest answers are

Iron and ferritin, and the number that matters more than "normal"

This is the single highest-yield test in this guide, particularly if you menstruate.

The trial worth knowing. A double blind randomised placebo controlled trial in 144 non-anaemic women aged 18 to 55 with unexplained fatigue gave either oral iron or placebo. Iron improved fatigue, and the subgroup analysis is the part that matters: **only women with ferritin concentrations at or below 50 micrograms per litre improved** (PMID 12763985). A 2026 double-blind placebo-controlled randomised trial in non-anaemic iron-deficient women of reproductive age found iron supplementation reduced fatigue scores and raised ferritin (PMID 41736325).

Why this changes the conversation with your doctor. Most laboratory reference ranges call a ferritin of 15 or 20 normal. Those trials found benefit up to a threshold of 50. So "your iron is normal" and "your iron is high enough to not be causing your fatigue" are different statements, and the first does not establish the second.

What to ask for. Ferritin, serum iron, TIBC, transferrin saturation, and a complete blood count, together. Ferritin alone is not enough because it rises with inflammation and can look reassuring in someone genuinely depleted, which is why the hs-CRP below is worth having alongside it.

About that hs-CRP. It is a nonspecific screening marker. A raised result says an inflammatory load exists somewhere. It does not say where it is coming from, and it never confirms a diagnosis. Here it earns its place mostly as the sanity check on your ferritin.

The caution. Iron is not a supplement to take speculatively. Unnecessary iron causes constipation and nausea, overload is a real risk, and iron is a leading cause of poisoning death in young children. Test, treat with a real dose, and retest.

B12, folate, and homocysteine

Deficiency causes fatigue plus neurological symptoms that get attributed to almost anything else. Ask for B12, RBC folate, and homocysteine together, because homocysteine rises when either is functionally low and can reveal a problem that individual levels in the low-normal range hide.

A specific trap: long-term acid suppression and metformin both reduce B12 absorption. If you take either and nobody has checked, ask.

Vitamin D and magnesium

Worth testing, worth correcting a genuine shortfall, and worth honest expectations. Correcting a real deficiency is a legitimate part of a fatigue workup. Neither is a stimulant, and neither will fix fatigue that is being driven by unrefreshing sleep or an unrecognized pacing problem.

Red blood cell magnesium is generally the more informative of the two magnesium options, since most magnesium is intracellular.

Key 6: the metabolic thermostat

Thyroid

Ask for **TSH with free T4 and free T3**, not TSH alone. TSH by itself is the standard screen and it can miss a conversion problem that free T3 shows.

I have deliberately not published a functional target range for thyroid markers in this guide, because I only publish ranges I can transcribe exactly from a validated source and thyroid is one where the right target depends on age, pregnancy status, and whether you are already on replacement. That is a conversation with a clinician looking at your numbers, not a table.

If antibodies have never been checked and your thyroid is abnormal, TPO and thyroglobulin antibodies answer whether it is autoimmune, which changes the outlook. The companion Hashimoto's kit covers that in full.

Blood sugar

Fasting glucose, HbA1c, and fasting insulin. Fasting insulin is the one that gets left off and it is often the most informative of the three, because insulin can climb for years while fasting glucose still reads normal. The pattern to look for is not diabetes, it is the energy crash that follows a high-carbohydrate meal and the 3pm collapse that resolves with food. Those are worth catching early, and the diet guide covers what to do about them.

Key 2: the HPA axis, stated more honestly than usual

This is where fatigue marketing is at its worst, so here is the careful version.

What is not established. “Adrenal fatigue,” as a diagnosis in which chronic stress exhausts the adrenal glands into underproduction, is not a recognized clinical diagnosis, and no panel can confirm it because there is nothing validated to confirm.

What the actual research shows, which is more interesting. A 2026 meta-analysis examined the neuroendocrine signature of myalgic encephalomyelitis and chronic fatigue syndrome and found evidence for a bioactive cortisol deficit alongside exaggerated feedback sensitivity, integrating neuroendocrine and psychological perspectives to describe chronic stress maladaptation in fatigue syndromes (PMID 42026257).

Read the difference carefully, because it matters. That is not “your adrenals are tired.” It is a subtler regulatory finding in a specific diagnosed population, and it does not translate into a supplement that fixes it. What it does establish is that the stress axis is genuinely involved, which makes the load side of this a clinical target rather than a lifestyle suggestion.

Where cortisol testing does earn its place, and where it does not. I do use four-point diurnal cortisol testing, salivary, in my practice. What I use it for is the shape of the curve across a day, because a flat and low pattern and a wired and high pattern both present as exhaustion and they call for different approaches. Knowing which one you are looking at changes what I do next, and a single morning blood draw cannot show it.

Be clear about what that is and is not. It is a pattern I find clinically useful, not a validated diagnostic threshold, and it does not diagnose “adrenal fatigue” because that diagnosis does not exist to be made. Anyone selling you a saliva kit as proof of a named adrenal condition, or as the justification for a glandular protocol, is going well beyond what the test can carry. Used for the pattern rather than for a label, it is worth running. Used as a diagnosis, it is not.

What I would actually do. If there is a clinical reason to suspect true adrenal insufficiency, which is a real and serious diagnosis, that is a morning cortisol and a physician. Otherwise, treat the stress axis through sleep, pacing, and load rather than through a panel.

Key 3 and Key 7: the two findings with no blood test

Sleep

Hours in bed are not hours of repair. Two things are worth ruling out before you conclude your fatigue is metabolic:

- **Sleep apnea.** If you snore, wake unrefreshed, wake with a headache, or someone has seen you stop breathing, ask for an assessment. Treating apnea treats the fatigue in a way no supplement will.
- **Insomnia as its own condition.** Cognitive behavioural therapy for insomnia has durable effects, with a systematic review and meta-analysis showing benefits on depressive symptoms sustained across the year following treatment (PMID 42435604), and fully automated digital versions have their own systematic review and meta-analysis (PMID 42240717). The digital option matters because access is the usual barrier.

Bring a two-week sleep log rather than an impression. Bedtime, wake time, estimated hours, and a refreshed-or-not rating.

Post-exertional malaise

The highest-consequence finding in this guide, and it is answered by a question rather than a test: **do your symptoms reliably worsen a day or two after exertion?**

If yes, that reframes everything. The supplement guide covers pacing, and the principle is that you plan to your average day rather than your best one.

Functional ranges for the markers in this guide

Clinical targets I use at Root Health, not validated diagnostic thresholds, narrower than the standard laboratory range on purpose.

Marker	Units	Functional target	Approximate standard range
Ferritin	ng/mL	50 to 150	25 to 300
Serum iron	ug/dL	65 to 110	40 to 150
TIBC	ug/dL	290 to 370	250 to 400
Transferrin saturation	%	20 to 40	15 to 45
Vitamin B12	pg/mL	500 to 900	200 to 1100
Folate (RBC)	ng/mL	600 to 1500	400 to 2000
Homocysteine	umol/L	5 to 7	under 10.0
Vitamin D (25-OH)	ng/mL	40 to 80	20 to 100
Magnesium (RBC)	mg/dL	5.2 to 6.8	4.0 to 7.2
Magnesium (serum)	mg/dL	2.0 to 2.5	1.5 to 2.7
Zinc (plasma)	ug/dL	70 to 110	60 to 120
Glucose, fasting	mg/dL	75 to 86	70 to 99
HbA1c	%	4.6 to 5.3	4.0 to 5.6
Fasting insulin	uIU/mL	2.6 to 5.0	2.0 to 10.0
hs-CRP	mg/L	under 0.5	under 1.0
White blood cells	K/uL	5.0 to 7.5	3.5 to 10.5
Albumin	g/dL	4.2 to 4.8	3.5 to 5.0

The ferritin row is the one to argue about. The trial evidence above found benefit in women below 50, which is inside most labs' normal range.

On homocysteine, lower is not better. The 5 to 7 target is a window, not a floor. A homocysteine driven below it points toward overmethylation and B vitamin intolerance, which is its own problem rather than a better result. An elevated result is a prompt to find the short input, B12, folate, B6, thyroid, or kidney function, not a number to push down as far as it will go.

Reading your results as a pattern

Pattern one: the depleted menstruating woman. Ferritin under 50, heavy periods, fatigue worse in the second half of the cycle. This is the most common correctable pattern in this guide and it has the best trial support (PMID 12763985).

Pattern two: unrefreshing sleep. Everything normal, eight hours in bed, wake exhausted. The answer is a sleep assessment, not a bigger nutrient panel.

Pattern three: post-exertional malaise. Symptoms crash a day after exertion. Pacing is the intervention, and the risk here is a well-meaning exercise prescription making things worse.

Pattern four: everything normal and none of the above. Common, and not a dead end. It moves the work to load, sleep regularity, blood sugar stability, and mood, none of which show on a panel and all of which are addressable.

The conversation with your doctor

A request list you can bring

- CBC with differential
- Ferritin, serum iron, TIBC, transferrin saturation
- Vitamin B12, RBC folate, homocysteine
- 25-hydroxy vitamin D
- TSH with free T4 and free T3
- Fasting glucose, HbA1c, and fasting insulin
- hs-CRP, understood as a nonspecific screen rather than a diagnosis
- Celiac serology, drawn while still eating gluten
- A sleep apnea assessment if you snore or wake unrefreshed

The sentence that helps

“My ferritin is X. I have read that trials of iron for unexplained fatigue in non-anaemic women found benefit below 50, and I would like to discuss whether that applies to me.” That is specific, it is accurate, and it opens a real conversation instead of a reassurance.

What these labs cannot tell you

- **Whether your mitochondria are working.** There is no validated clinical test, and anyone selling one is ahead of the evidence.
- **Whether you have adrenal fatigue.** Not a recognized diagnosis, and a salivary panel does not establish it.
- **Whether you have post-exertional malaise.** That is your history, and it matters more than the panel.
- **Why you are tired, if everything is normal.** A clean panel is information, not dismissal.

When to seek clinical review

Promptly: unintentional weight loss, fever or night sweats, a new lump, bleeding, chest pain, breathlessness, fainting, new confusion. New or steadily worsening fatigue over 60.

Soon: fatigue lasting more than six weeks without explanation, snoring or witnessed pauses in breathing, fatigue with low mood or loss of interest, or fatigue that reliably worsens a day after exertion.

Now: thoughts of harming yourself. Call or text 988.

Your next step

The companion guides do the rest: **The Energy Diet Guide** covers blood sugar and the gut, and **The Energy Supplement Guide** covers what has actually been studied, plus the pacing and sleep work that outperforms most of it.

> ***Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=energy-labs***

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

> ***Get in touch: rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=energy-labs***

If you would rather start on your own, start tracking. In fatigue, the pattern across weeks is the diagnosis:

> ***Free symptom tracker: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=energy-labs***

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Educational content only. Not medical advice. Not a diagnosis. Not a substitute for your clinician. Persistent or unexplained fatigue deserves a medical workup.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

See a doctor before dieting your way through fatigue if it comes with unintentional weight loss, fever or night sweats, a new lump, bleeding, chest pain, breathlessness, fainting, or new confusion, or if it began after age 60 or has worsened steadily over weeks.

If you are pregnant, nursing, underweight, diabetic, or have a history of disordered eating, talk with a clinician before restricting anything. In a fatigue population, an unnecessary restriction costs more than it does in most conditions, for the reason in the next section.

If you have thoughts of harming yourself, call or text 988.

The thing most fatigue diet advice gets wrong

Cooking is an activity. Shopping is an activity. Standing at a counter chopping vegetables for forty minutes is an activity, and it comes out of the same daily energy budget as everything else you need to do.

Almost every fatigue diet on the internet is written as though the reader has normal energy to spend on implementing it. Then it prescribes an elimination protocol with three home-cooked meals a day, and the person following it spends their entire envelope on food preparation and crashes.

So the first rule of this guide is that the plan has to cost less energy than it returns. Everything below is written to that constraint, and where a strategy is expensive to run, I say so.

If your symptoms reliably worsen a day or two after exertion, that is post-exertional malaise, and this constraint is not a nicety. Read the pacing section of the supplement guide first.

Where food sits among the eight keys

Unlock Your Energy works through eight root causes: cellular, HPA axis, sleep, gut, nutrients, thyroid and blood sugar, pacing, and testing. The lab guide maps all eight and tells you which are yours.

Food is a direct lever on two of them, the nutrient key and the blood sugar half of the metabolic key, and an indirect one on the gut. It is not a lever on sleep apnea, on post-exertional malaise, or on a thyroid that needs replacing. Knowing which of your keys food can actually reach is what stops you spending a year on the wrong one.

Part one: blood sugar, the most reachable win

The 3pm collapse that resolves the moment you eat is not a personality trait. It is the most fixable pattern in this guide, and the fix is mechanical.

What to do:

- **Eat protein at breakfast.** The single change that most reliably flattens the rest of the day.
- **Do not go more than four or five hours without food** during your waking day.
- **Pair carbohydrate with protein, fat, or fiber** rather than eating it alone. A slice of toast alone and toast with eggs behave differently.
- **Watch liquid sugar specifically**, including the coffee drink that is functionally a dessert.
- **Notice what happens ninety minutes after you eat.** That is your data.

Sourcing note, stated plainly: the blood sugar guidance above is standard clinical nutrition and physiology rather than a finding from a trial in fatigue patients. I have not attached a citation to it because I did not find one that studied this pattern in this population. It is included because it is low risk, cheap, and reversible, and because the pattern is easy to test on yourself in a week.

Part two: eating for the iron finding

This is the part of the diet guide with actual trial evidence behind it, and it connects directly to the lab guide.

Why it matters. In a double blind randomised placebo controlled trial of 144 non-anaemic women with unexplained fatigue, iron improved fatigue, and the benefit was confined to women with **ferritin at or below 50** (PMID 12763985). A 2026 randomised placebo-controlled trial in non-anaemic iron-deficient women likewise found reduced fatigue and increased ferritin with supplementation (PMID 41736325).

What that means at the table. If your ferritin is low, diet supports repletion but rarely achieves it alone, so this section sits alongside the supplement conversation rather than replacing it.

Absorption is where the practical gains are:

- **Heme iron**, from red meat, poultry, and fish, absorbs far better than the non-heme iron in plants.
- **Vitamin C with the meal** improves non-heme absorption. Peppers, citrus, tomato, or strawberries alongside beans, lentils, or greens.
- **Tea and coffee block it.** Separate them from iron-rich meals and from any iron supplement by a couple of hours. This one change is free and it is the most common unforced error.
- **Calcium competes.** Do not take a calcium supplement at the same time as iron.
- **Cast iron cookware** contributes a small amount, which is not a treatment but is not nothing.

Part three: the gut key, without overclaiming it

The gut sits in Key 4 of the ebook, and it earns its place for two concrete reasons rather than a general terrain argument.

Absorption. You cannot correct an iron or B12 deficiency through a gut that is not absorbing. That is the practical link, and it is why the lab guide asks for celiac serology.

Get celiac testing before you remove gluten. Celiac disease presents with fatigue often enough that it belongs in any fatigue workup, and the serology is only interpretable while you are still eating gluten. Remove it first and you lose the ability to get a clean answer without a deliberate gluten challenge later. This is the single most common way this diagnosis gets missed for years.

The bidirectional part. Sleep disruption and the gut interact, and a 2026 review of sleep disruption and the gut-brain dialogue describes that bidirectional communication while being explicit that mechanistic and translational work is still needed (PMID 42092445). Directional rather than settled, and it points the same way as the sleep work in the supplement guide.

What I am not going to claim. That a gut protocol resolves fatigue. There is no trial supporting that, and in a population prone to spending scarce energy on complicated regimens, overclaiming here does real harm.

Part four: a plan that costs less energy than it returns

This is the section I would keep if I could only keep one.

Cook once, eat three times. Batch on your best day of the week, portion immediately, freeze. Not leftovers in the fridge for five days, actual frozen portions.

Keep a no-cook tier stocked at all times, for the days when cooking is not happening. Not as a failure plan, as a normal part of the system: tinned fish, pre-cooked rice, frozen vegetables, eggs, nut butter, plain yogurt, fruit.

Lower the bar for what counts as a meal. Protein plus a vegetable plus a carbohydrate is a meal. It does not need to be assembled or attractive.

Use the shortcuts without guilt. Pre-chopped vegetables, frozen produce, rotisserie chicken, and grocery delivery all convert money into energy, which is the correct direction of trade in this condition.

Sit down to eat. Standing at the counter costs you more than you think, and digestion runs better when you are not rushing.

A sample rhythm rather than a menu. Batch two proteins and two vegetables on your best day. Cook a grain that keeps. Eat those in rotation for three days, then use the no-cook tier for two, then repeat. That is the whole system.

What I deliberately left out, and why

- **Elimination diets, unless there is a specific reason.** They are expensive to run in energy terms, they narrow nutrition, and in an unexplained-fatigue population the yield is low. Celiac testing is the exception, and it is a blood test rather than a diet.
- **Extended fasting and aggressive intermittent fasting.** Skipping meals is the opposite of the blood sugar work above, and in a fatigue population it commonly backfires.
- **Ketogenic diets for fatigue.** Demanding to run, and I did not find trial evidence supporting them for unexplained fatigue.
- **"Energy" drinks, shots, and stimulant blends.** They borrow energy against later, and where post-exertional malaise is present, feeling able to do more than your envelope allows is actively harmful.
- **Detox and cleanse protocols.** No fatigue outcome evidence, real energy cost.
- **Green powders as a substitute for testing.** If your ferritin is 18, a scoop of powder is not the answer, and it will delay the one that is.

Telling you what to skip is part of what you paid for, and in this condition it is also giving you your afternoon back.

Running a fair experiment

Weeks 1 and 2, baseline. Change nothing. Score energy out of ten daily and mark crash days.

Weeks 3 to 6, one change. Protein at breakfast is the one I would start with, because it is the cheapest to implement.

Week 7, judge it. Compare against baseline. If nothing moved, the lever is elsewhere, and the lab guide tells you where to look.

Then one more change. Never two at once.

Realistic expectations

What is realistic. A steadier afternoon within two weeks of fixing breakfast. Meaningfully better energy over two to three months if a real iron or B12 deficiency gets corrected.

What is not realistic. Diet resolving fatigue driven by untreated sleep apnea, post-exertional malaise, an unreplaced thyroid, or depression. Those need their own treatment.

And one thing worth saying plainly. If you have been told your fatigue is because you should eat better, and you have already been eating well for a year, the problem is not your willpower and it is not your diet. Go back to the lab guide.

When to seek professional care

Promptly: unintentional weight loss, fever or night sweats, a new lump, bleeding, chest pain, breathlessness, fainting, new confusion, or new fatigue over age 60.

Soon: unexplained fatigue beyond six weeks, snoring or witnessed pauses in breathing, fatigue with low mood or loss of interest, a diet that has narrowed below roughly twenty foods, or a reliable crash a day after exertion.

Now: thoughts of harming yourself. Call or text 988.

Working with Root Health

This guide is written to be safe and useful for a reader I have never met. A personalized plan starts from your labs, your history, and whether post-exertional malaise is part of your picture.

> ***Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=energy-diet***

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

> ***Get in touch: rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=energy-diet***

If you would rather start on your own, track first. In fatigue the pattern across weeks is the diagnosis:

> ***Free symptom tracker: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=energy-diet***

Read the companion documents alongside this one. **Understanding Your Energy Labs** covers the ferritin threshold worth arguing about, and **The Energy Supplement Guide** covers pacing and sleep, which outperform most of what you can buy.

References

All references were verified against live PubMed records on 2026-08-09, including a publication type and corrections check for retractions and errata. Verification detail is documented in `Energy-Kit-Citation-Ledger.md`.

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Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

See a doctor before treating fatigue yourself if it comes with unintentional weight loss, fever or night sweats, a new lump, bleeding, chest pain, breathlessness, fainting, or new confusion, or if it started after age 60 or has worsened steadily over weeks. Fatigue is a symptom of many things and some of them need finding.

If you have thoughts of harming yourself, call or text 988 in the United States, or go to an emergency department.

Supplements in the United States are regulated as food, not as drugs. Quality varies. Look for independent third-party testing.

Why nutrient evidence looks thinner than it is

A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back. You cannot patent magnesium, an herb, or a food. So the evidence hierarchy this guide grades against is not neutral: it systematically over-documents patentable drugs and under-documents nearly everything else. That is measurable, not rhetorical. A Cochrane methodology review found that manufacturer sponsorship of drug and device studies produces more favourable results and conclusions than independent funding, through a bias that standard quality checks do not catch (PMID 28207928).

Two honest consequences follow, and they pull in opposite directions.

First, "no evidence" has three different meanings, and this guide tries to always tell you which one applies. *Tested and failed*: a real trial ran and the answer was no. *Never properly tested*: nobody could afford the trial, so absence of evidence is not evidence of absence. *Tested small and won*: real but modest trials, usually measuring markers rather than outcomes, because markers are what a small budget can afford.

Second, the asymmetry earns nutrients humility, not automatic credit. When isolated nutrients have received large publicly funded trials, the results have been mixed: some genuine wins, some nulls, and some harms. Small nutrient trials also carry their own publication bias toward positive results, just as drug trials tilt toward their sponsor. So this guide applies one rule in both directions: name the funding, name the evidence tier, and let you see both.

The safety line that comes before everything else

If your symptoms reliably worsen a day or two after exertion, do not let anyone talk you into pushing through.

That pattern is post-exertional malaise, and it is the defining feature of myalgic encephalomyelitis and chronic fatigue syndrome. Where it is present, graded exercise is not the answer and can make you worse. This is Key 7 of *Unlock Your Energy* and it is the most important thing in this kit.

Everything below is written to work alongside pacing rather than against it.

Two foundations that beat everything you can buy

I would rather you spend your effort well than spend money here, so these come first.

Pacing, if post-exertional malaise is part of your picture

A scoping review of pacing, the energy-management approach used in ME/CFS, describes it as a prominent coping strategy while stating plainly that the literature base is not yet sufficient to dictate treatment practice (PMID 37838675). Co-production work with adults living with ME/CFS puts energy management at the center of daily life, alongside sleep, diet, and psychological support (PMID 39961545). Post-treatment and post-infectious fatigue syndromes are increasingly recognized as a group, with calls for better research and inclusive trial design (PMID 41967005).

The honest state of it: pacing is the consensus practical approach and the trial base behind it is thinner than it should be. It is also the approach least likely to harm you, which matters more than usual in a condition where the standard alternative caused harm.

What it looks like.

- Find your envelope on an **average** day, not your best one, and plan to the average.
- Break activity into short blocks with rest between, rather than one push followed by a crash.
- Stop before you are empty. The instinct to finish the task is the thing that costs you tomorrow.
- Expect a non-linear course and judge in months.

Sleep, treated as a treatment

Hours in bed are not hours of repair. Two things outrank every capsule below.

Rule out sleep apnea if you snore, wake unrefreshed, wake with headaches, or someone has seen you stop breathing.

And if the problem is insomnia, the treatment has real evidence. Cognitive behavioural therapy for insomnia produces effects on depressive symptoms sustained across the year after treatment in a systematic review and meta-analysis (PMID 42435604), and fully automated digital delivery has its own systematic review and meta-analysis (PMID 42240717). The digital versions matter because access is usually the barrier, and they cost a fraction of a supplement stack.

How to read this guide

Test before you supplement, wherever a test exists. In fatigue this is not a formality, because the two nutrients with the best evidence only helped the people who were actually low.

Give it eight to twelve weeks, and track two numbers most days: energy out of ten, and whether you crashed after exertion.

One change at a time. Fatigue fluctuates on its own, and a stack teaches you nothing.

Tier 1: correct what is measurably low

Iron, only if your ferritin says so

The trial that defines this entry. In a double blind randomised placebo controlled trial of 144 non-anaemic women aged 18 to 55 with unexplained fatigue, oral iron improved fatigue, and the subgroup analysis found that **only women with ferritin at or below 50 micrograms per litre improved** (PMID 12763985). A 2026 double-blind placebo-controlled trial in non-anaemic iron-deficient women of reproductive age likewise found reduced fatigue scores and increased ferritin with iron supplementation (PMID 41736325).

How to read that. This is one of the cleanest findings in the whole fatigue literature, and it comes with a built-in filter: it worked in the depleted, and it did not work in everyone else. So the test comes first.

Typical studied range. 80 mg daily of elemental iron as ferrous sulphate in the 2003 trial (PMID 12763985). Dose and duration belong with your clinician, and a retest belongs at the end.

Cautions. Constipation and nausea are common and are the usual reason people quit. Iron overload is a real risk in people who do not need it. Separate iron from thyroid medication by four hours. **Keep it locked away from children**, since it is a leading cause of childhood poisoning death.

B12, folate, vitamin D, and magnesium, on the same principle

Correct a documented shortfall. Do not supplement speculatively and expect an energy effect, because that is not what the evidence supports. The lab guide covers what to ask for and what the functional targets are.

Two practical notes. Long-term acid suppression and metformin both reduce B12 absorption, so if you take either, ask for a level. And magnesium in the evening is well tolerated and may help sleep, which in this condition is a reasonable secondary reason to take it.

Tier 2: mitochondrial support, with honest expectations

Coenzyme Q10, with or without NADH

What the research shows. A prospective, randomized, double-blind, placebo-controlled trial tested CoQ10 plus NADH in people with myalgic encephalomyelitis and chronic fatigue syndrome, and reported benefit for perceived cognitive fatigue and health-related quality of life, describing the combination as generally safe while calling for further work to confirm the clinical benefits and clarify mechanism (PMID 34444817). A separate randomized, double-blinded, placebo-controlled study of ubiquinol-10 at 150 mg daily over 12 weeks in chronic fatigue syndrome found improvement in several symptoms, including measures of autonomic and cognitive function (PMID 27125909).

How to read it. Two small placebo-controlled trials in a diagnosed population, both positive, neither large. That is better than most things sold for fatigue and well short of settled. If you try it, try it for the right reason: modest improvement in perceived fatigue and cognition, not a return to your old energy.

Typical studied range. 150 mg daily of ubiquinol-10 for 12 weeks in the trial above (PMID 27125909). The combination trial used CoQ10 with NADH (PMID 34444817).

Timing. With a fat-containing meal, since absorption depends on it. Earlier in the day, as some find it activating.

Cautions. CoQ10 can reduce the effect of warfarin, which requires prescriber involvement. It may modestly lower blood pressure and blood sugar. Not established in pregnancy or nursing.

Tier 3: ask your prescriber first

- **Iron**, always, because it should follow a test and a dose decision rather than a guess.
- **Anything stimulant-like**, including high-dose caffeine products and “adrenal support” blends containing glandulars or licorice, particularly if you have high blood pressure. In a fatigue population these borrow energy rather than create it, and in post-exertional malaise they invite exactly the overexertion that causes the crash.
- **Thyroid glandulars and iodine**, which are not benign in autoimmune thyroid disease and are covered in the Hashimoto’s kit.
- **Anything at all in pregnancy or nursing.**

What I deliberately left out, and why

- **Adrenal glandular and adaptogen protocols sold on the back of an "adrenal fatigue" diagnosis.** That diagnosis is not recognized, and the real neuroendocrine finding in ME/CFS is subtler than the marketing, involving a bioactive cortisol deficit and exaggerated feedback sensitivity on meta-analysis (PMID 42026257). It does not translate into a product that fixes it. Note the distinction from the testing itself: I do use four-point diurnal cortisol clinically, for the shape of the curve rather than to name a condition, and the lab guide in this kit explains exactly what that does and does not tell you.
- **NAD precursors sold for fatigue.** Interesting biology, expensive, and the fatigue trial evidence does not yet justify the price. Note that the CoQ10 trial above used NADH, which is a different compound from the NMN and NR products being marketed.
- **Mitochondrial "support" stacks with a dozen ingredients at token doses.** You cannot tell what worked, and the doses that were studied are specific.
- **B12 injections in people whose B12 is normal.** Popular, and not supported.
- **Anything that promises energy quickly.** In a condition where overexertion causes crashes, a product that makes you feel able to do more than your envelope allows is not a good outcome.

Telling you what to skip is part of what you paid for.

Running a fair trial

Weeks 1 and 2, baseline. Change nothing. Score energy out of ten daily, and mark any day you crashed after exertion.

Weeks 3 to 14, one intervention. Start with whichever deficiency your labs actually showed. If nothing was low, start with the sleep work rather than a bottle.

Week 15, judge it honestly. Compare your last two weeks against baseline. In this condition, “fewer crashes” is often a better outcome signal than “more energy,” and it moves first.

Bring the tracker to appointments. Three months of daily scores is more useful than any description from memory.

Realistic expectations

What is realistic. Correcting a real iron or B12 deficiency can produce a genuine, noticeable change over two to three months. Everything else here is modest.

What is not realistic. A supplement resolving fatigue whose cause is unrefreshing sleep, an unrecognized pacing problem, untreated apnea, or depression. Those need their own treatment, and no amount of mitochondrial support substitutes.

Timeline. Eight to twelve weeks minimum. Iron often takes longer, because stores refill slowly.

Cost sense. If something has done nothing after twelve weeks at a sensible dose, stop paying for it.

When to seek professional care

Promptly: unintentional weight loss, fever or night sweats, a new lump, bleeding, chest pain, breathlessness, fainting, new confusion, or new fatigue over age 60.

Soon: unexplained fatigue beyond six weeks, snoring or witnessed apneas, fatigue with low mood or loss of interest, or a reliable crash a day after exertion.

Now: thoughts of harming yourself. Call or text 988.

Where to find quality versions of these

Nothing here names a brand, on purpose. The evidence supports ingredients and forms.

Look for third-party testing. NSF, USP Verified, or equivalent.

Match the form to the trial. Ubiquinol-10 at 150 mg is what was studied, and it is not the same as a generic CoQ10 at an unstated dose.

Read the whole panel. Fatigue blends frequently hide caffeine or other stimulants, which is the last thing you want if crashes are your problem.

If you would rather not evaluate all of that yourself, I keep a screened dispensary through Fullscript, which ships directly to you. I hold no inventory myself, and everything in it is screened against the standards above.

> ***Root Health dispensary: us.fullscript.com/welcome/drseay***
>
> *Disclosure: Root Health earns a portion of dispensary purchases. That revenue supports the practice, and it is also why this guide was written to be complete and usable on its own. Nothing here was included, excluded, ranked, or dosed based on what it earns, and the two highest-value items in this guide, pacing and sleep, cost nothing at all.*

Working with Root Health

Everything here is general education, written to be safe for a reader I have never examined. A personalized plan starts from your labs, your history, and whether post-exertional malaise is part of your picture, because that single question changes the whole plan.

> ***Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=energy-supplement***

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

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Read the companion documents alongside this one. **Understanding Your Energy Labs** covers what to test and the ferritin threshold worth arguing about, and **The Energy Diet Guide** covers blood sugar and the gut.

References

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